

BridgeFinder: Finding People Across Structural Holes

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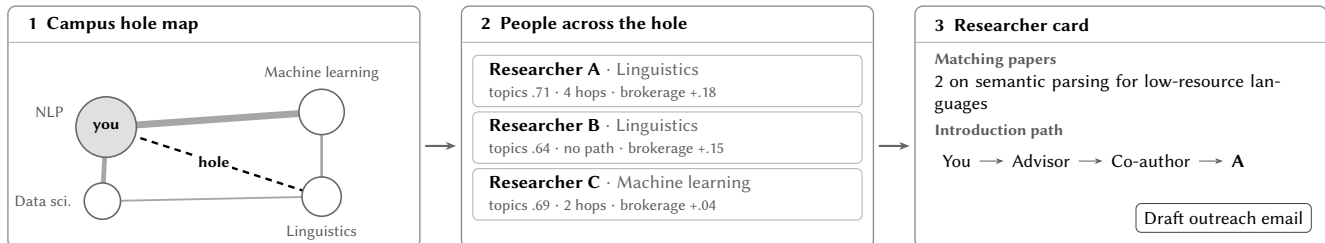


Figure 1: Core flow. (1) UIUC co-authorship communities; dashed = a structural hole (similar topics, almost no joint papers). (2) Researchers across the hole, ranked. (3) Their matching papers and an introduction path.

1 OBJECTIVE

Students seeking a research mentor, and researchers seeking collaborators, search through people they already know: an advisor's contacts, their lab, their department. That returns people who are already close. Yet new information travels through weak ties [5], and people who broker between groups that don't talk have better ideas [4]. At UIUC ([N] researchers, [M] units), two groups can work on the same problem for years without a joint paper. **BridgeFinder** finds these gaps in the co-authorship graph and names the people on the other side.

2 RELATED WORK

Illinois Experts [1] and Google Scholar search researchers by keyword but ignore network position: your co-author's co-author ranks like a stranger. LinkedIn's "People You May Know" and Twitter's Who-to-Follow [6] use triadic closure, the opposite of brokerage. Link prediction [7] finds *likely* ties; we rank ties by *value*, using Burt's constraint [3].

3 COURSE CONCEPTS & NETWORK ANALYSIS

Core concept: **structural holes and brokerage** (Lesson 1), with **community discovery** (Lesson 10) and **weak ties** and **decentralized search** (Lessons 1, 13).

Graph and method. Nodes are UIUC researchers in OpenAlex [9]. Edges are 2015–2024 co-authorships weighted by joint papers. Each node has a topic vector. (1) Louvain modularity [2] finds communities for the map. (2) Two communities form a *hole* when their topics are similar but few edges join them. (3) For a user u , each researcher v is scored by $\text{topic similarity} \times \text{graph distance} \times \text{brokerage gain}$, the drop in u 's constraint if the tie $u-v$ were added. (4) A weak-tie shortest path offers an introduction.

Alternatives from class. Triadic closure only finds people already near you; topic similarity alone can't tell a stranger from a friend of a friend. Girvan–Newman [8] could not split a 150-node graph in 148 s in our pilot; Louvain split 610 nodes in 1.5 s.

Claim that could be false. From the 2015–19 graph, BridgeFinder ranks the actual partners of new *cross-community* collaborations

(2020–24) higher than triadic-closure or topic-only ranking, and those collaborations' papers are cited more than new within-community ones in the same fields (Burt's claim).

4 PROPOSED WORK

A web app for Figure 1 (Python/networkx over an OpenAlex snapshot) plus the backtest. **Oct 30:** graph, holes, backtest, prototype, feedback round 1. **Mid-Nov.:** person cards, intro paths, round 2. **Dec 9:** round 3, evaluation. *Assumptions:* OpenAlex disambiguation is good enough (we will hand-check 50 authors); topic vectors capture "similar research"; students without papers are placed through their advisor or lab; public data and drafted (never auto-sent) emails keep outreach appropriate.

5 PRELIMINARY DESIGN

In Figure 1, the map comes first, so users see *where* the holes are before seeing names. Every suggestion shows its reasons, because unexplained recommendations are hard to trust.

6 USERS

UIUC students seeking mentors or collaborators outside their lab, plus a few faculty seeking interdisciplinary partners. We will recruit 10–15 students per round, for three rounds, via ACM@UIUC SIGs, the Office of Undergraduate Research and our labs. Each rates suggestions for their own interests.

7 EVALUATION

(1) **Backtest recall@k** of new cross-community collaborators vs. triadic closure, topic-only and degree. (2) **Relevant-and-new rate** of suggestions vs. triadic-closure and topic-only lists (within-subject). (3) **Outreach intent**, with a follow-up on emails sent. Success means beating triadic closure on (1) and (2); we will report where we do not.

Use of AI: Claude (Anthropic) helped brainstorm, prototype and draft; we checked everything.

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